# VINAY K R

**Applied AI Systems Engineer | Generative AI & Backend Architecture**

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## PROFESSIONAL SUMMARY


Applied AI and backend systems engineer with nearly three years of production experience building resilient distributed services, autonomous agent workflows, and low-latency retrieval pipelines. Specialized in Python/FastAPI, LangGraph multi-agent orchestration, hybrid RAG (BM25 + dense semantic hashing with Reciprocal Rank Fusion), structured LLM generation, and automated evaluation harnesses. Combines enterprise software foundations in transaction safety, streaming architectures, and Redis caching with end-to-end LLM observability (OpenTelemetry).


## TECHNICAL SKILLS

Applied AI & Agent Systems: Python, FastAPI, LangGraph, Multi-Agent Orchestration, OpenAI/Anthropic APIs, Structured Outputs, Hybrid RAG, BM25, Dense Semantic Hashing, Reciprocal Rank Fusion (RRF), Redis Vector Caching

Evaluation & Telemetry: Reproducible Benchmarks, Recall@k, MRR, Calibration & Abstention, OpenTelemetry, Trace/Span Modeling, Prompt Injection Defense, Hallucination Guardrails, pytest, Mypy, Ruff

Backend & Distributed Systems: Go, PostgreSQL/PostGIS, SQL, Redis Pub/Sub, Docker, CI/CD, Async Workers, WebSockets, SSE Streaming, Pydantic, Authentication, RBAC

Enterprise Engineering: TypeScript, Node.js, Angular, C#/.NET, SQL Server, Event-Driven Architecture, Unit/Integration Testing, Production Telemetry

## PROFESSIONAL EXPERIENCE

### Applied AI & Backend Software Engineer | Liminal Custody

Nov 2025 – Mar 2026 | Bengaluru

- Architected automated policy verification and compliance audit workflows using FastAPI, TypeScript, and Redis semantic caching, reducing manual custody approvals.
- Engineered real-time anomaly detection and risk evaluation pipelines with atomic rollback safeguards, protecting digital asset operations against anomalous states.
- Implemented organization-scoped RBAC, step-up authentication, and OpenTelemetry distributed tracing across async custody micro-services, ensuring enterprise auditability.

### AI / Backend Software Engineer | Light & Wonder

Aug 2023 – Jul 2025 | Bengaluru | Promoted from Associate Software Engineer

- Engineered machine learning recommendation pipelines and player personalization models using predictive algorithms and historical engagement telemetry.
- Designed high-throughput real-time event streaming systems over WebSockets and C#/.NET APIs, integrating telemetry pipelines to track system latency and model inference performance.
- Developed comprehensive automated regression test suites and diagnostic tooling; resolved asynchronous subscription lifecycle leaks in enterprise gaming hardware.

### ML & Full-Stack Intern | Light & Wonder

Mar 2023 – Jul 2023

- Developed an ML-driven game recommendation application utilizing collaborative filtering, user preference scoring, and explicit rating feedback loops.

## PROJECTS

### FastAPI GenAI Agent Patterns | Flagship Applied AI Reference Implementation | Aug 2026

github.com/vi-nayKR/fastapi-genai-agent-patterns

- Architected typed LangGraph multi-agent workflow featuring deterministic supervisor routing, structured Pydantic outputs, checkpointed human-in-the-loop approval, and SSE event streaming.
- Engineered versioned hybrid retrieval combining Okapi BM25 and dense semantic hashing via Reciprocal Rank Fusion (RRF k=60), achieving 100% Recall@3 and 0.9545 MRR across tenant-isolated corpora.
- Built a 30-case reproducible evaluation benchmark measuring hallucination abstention, citation precision, and prompt-injection defense with zero unauthorized mutations; 49 tests passing with strict Mypy.

### Medha Platform API | Modular Go Backend & Intelligent Geospatial Dispatch | 2026

github.com/vi-nayKR/medha-platform-api

- Engineered 21-domain modular backend monolith featuring PostGIS proximity dispatch (ST_DWithin), Redis 8 Pub/Sub WebSockets, and cross-context atomic transaction propagation.
- Implemented automated provider-client matching state machine with async lead expiration workers, strict RFC 7807 error contracts, and complete zero-regression race-condition test coverage.

## PUBLICATIONS

### Data Visualisation of Time-Tradable Assets Using Machine Learning | IEEE Xplore | 2023

ieeexplore.ieee.org/document/10275962

- Co-authored IEEE-published research (DOI: 10.1109/NCNSP56992.2023.10275962) benchmarking ML architectures (Linear Regression, Decision Tree, SVR, PyTorch LSTM) on financial time-series forecasting.
- Built interactive analytics platform with FastAPI and Angular to visualize model error metrics (MSE/RMSE), technical indicators, and OHLCV market features with low-latency caching.

## EDUCATION

B.E., Computer Science & Engineering | Siddaganga Institute of Technology | 2023 | CGPA: 8.65/10